UNIVERSITY OF RWANDA

College Of Science and Technology

Computer Engineering

Year *3*

Gako Campus

FLUTTER MOBILE DEVELOPMENT

Lab 6

Reg n°: **223026685**

223026674

2230267475

Names: Jerome HAKIZIMANA

Lewis NGENDAHIMANA

Clement NZABAMWITA

Push Notifications with Firebase Cloud Messaging (FCM)

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1. Objective

The objective of this lab was to create a Flutter mobile application capable of receiving push notifications from Firebase Cloud Messaging (FCM). The application was built as part of the my_devices project which covers multiple device feature topics from Lesson 8.

The lab required the following:

1. *Integrating Firebase Cloud Messaging (FCM) into the Flutter app*
2. *Requesting notification permission from the user*

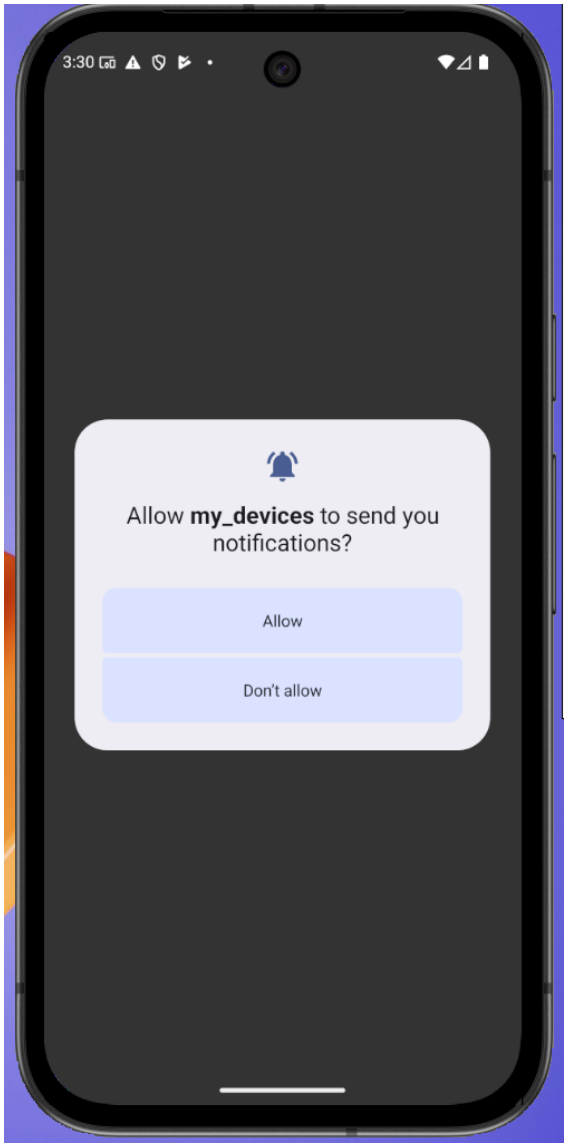

Figure 12. Requesting notification permission from the user

3. *Retrieving and displaying the FCM device token on screen*
4. *Receiving push notifications sent from Firebase Console*
5. *Showing a popup dialog when a notification arrives in the foreground*
6. *Displaying message history inside the app UI*

7. *Testing notifications while the app is open and in the background*

2. Screenshots

Screenshot 1 — Application Home Screen

This screenshot shows the main home screen of the my_devices application. It displays four feature cards, each navigating to a different device feature screen covered in Lesson 8. The app bar shows the title "My Devices — Lesson 8" with a deep purple theme.

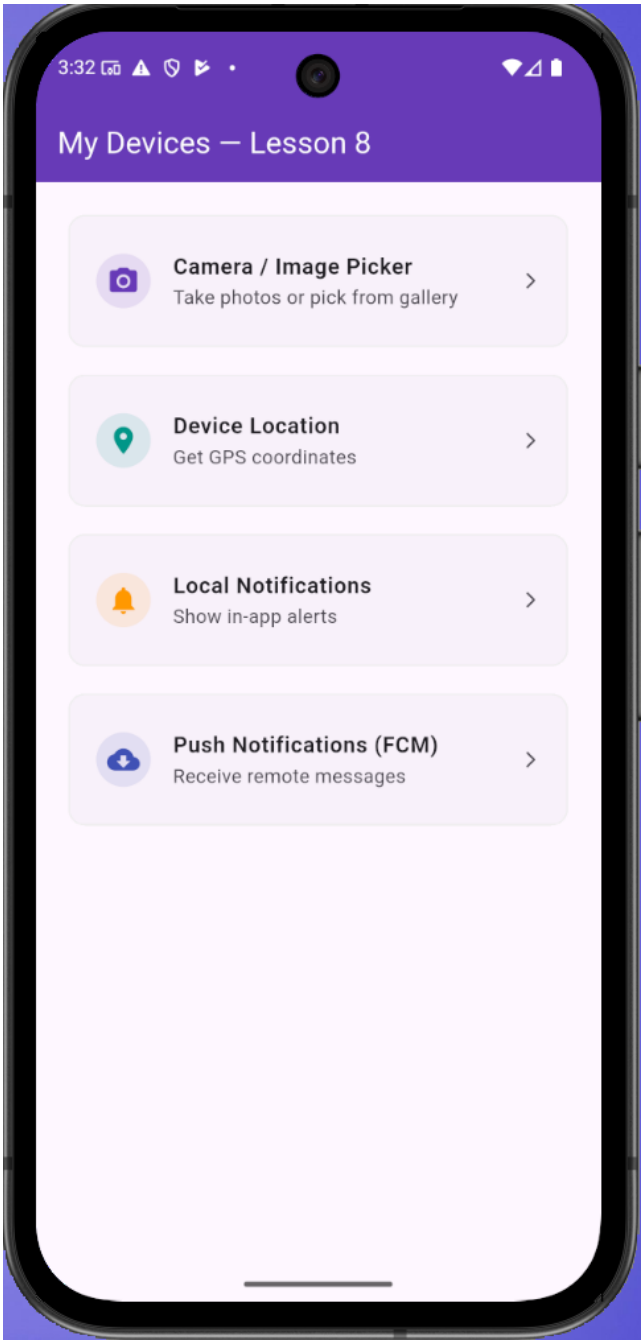

Fig. 1 — Home screen showing the 4 device feature cards

Screenshot 2 — Permission Granted (Console Log)

This screenshot from the Flutter console log confirms that the app successfully requested and received notification permission from the user. It also shows the FCM token being printed to the console, which is the unique identifier for the device used to send targeted notifications from Firebase Console.

```
I/flutter (17604): ✅ Permission granted for notifications
I/flutter (17604): 📱 FCM Token: eGJTYec2SPSkwCrHNl4GnC:APA91bEnskBczvY9te7Cl39AQZVjvGM_vSuZyUxMS1jhZ9S3ZyASL0tfHS37GRrFLsUE2rkY5-7eJzhiBC3Ki7K-x
u4zylxZMtu-6VJxmKxbajmCqHu5GSY
D/ActivityThread(17604): Package [com.google.android.apps.docs] reported as REPLACED, but missing application info. Assuming REMOVED.
D/ActivityThread(17604): Package [com.google.android.apps.docs] reported as REPLACED, but missing application info. Assuming REMOVED.
I/flutter (17604): ✅ Subscribed to topic: all_devices
```

Fig. 2 — Console log confirming permission granted and FCM token printed

Screenshot 3 — FCM Token Displayed on Screen

This screenshot shows the Push Notifications screen displaying the FCM token on screen. The token is shown in a selectable text field with a "Copy Token" button below it. The green banner at the top confirms that notifications are enabled. The "How to test" instructions guide the user through the process of sending a test notification from Firebase Console.

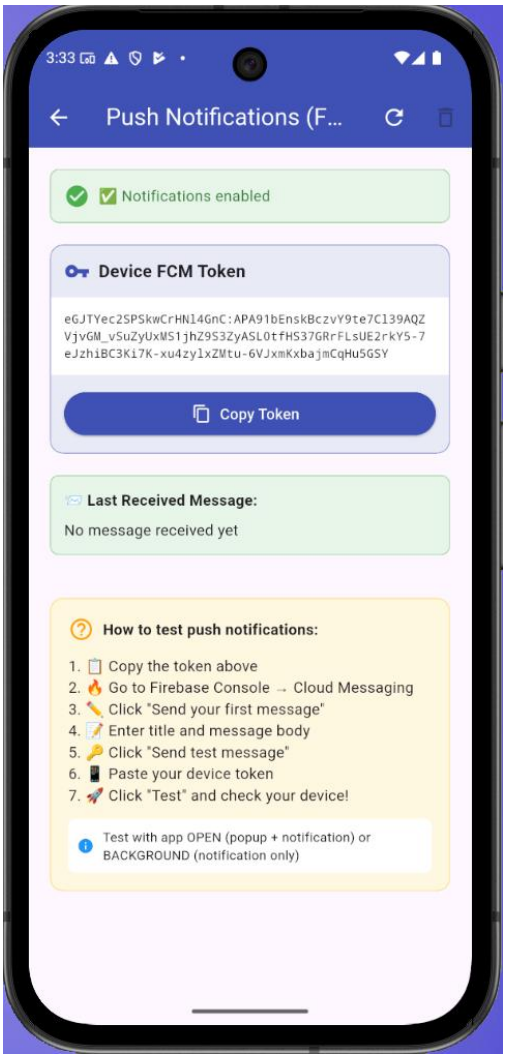

Fig. 3 — Push Notifications screen showing the FCM token and copy button

Screenshot 4 — Token Entry in Firebase Console

This screenshot shows the Firebase Console "Test on device" dialog. The FCM token was pasted into the registration token field and is shown as checked in the "Recently Used" section. Clicking the Test button sends a notification directly to the emulator using that token.

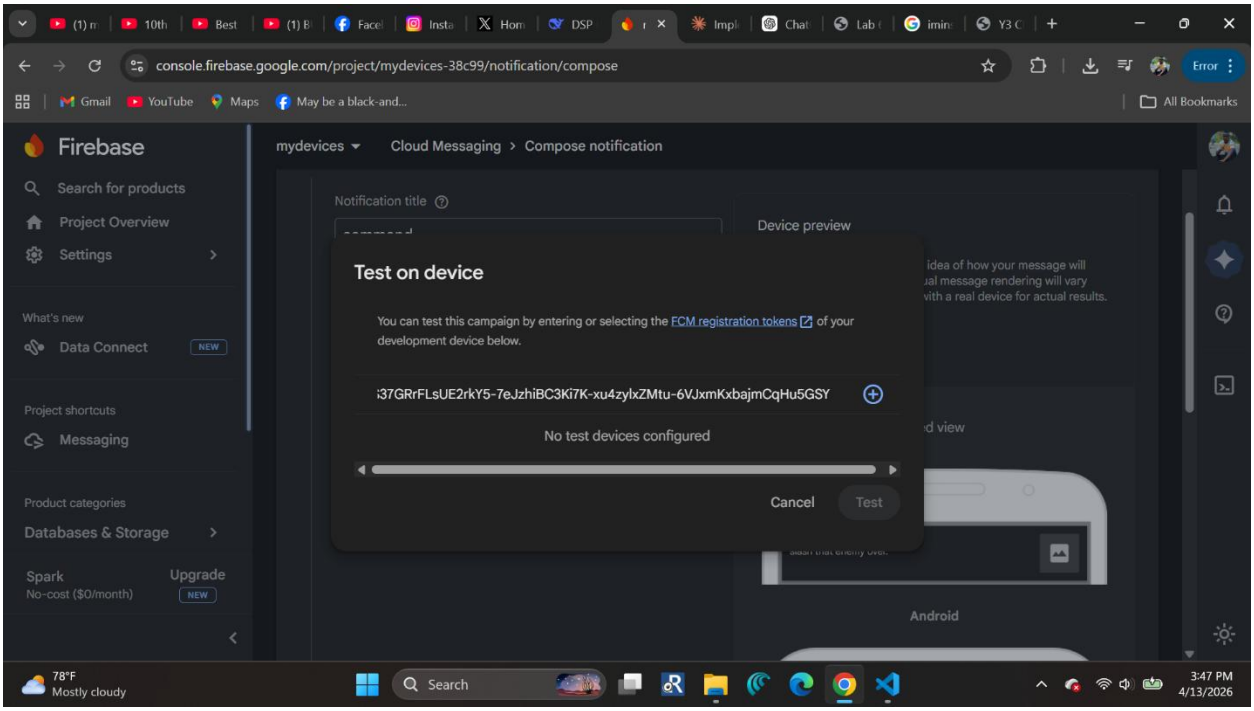

Fig. 4 — Firebase Console showing token entered and selected for testing

Screenshot 5 — Composing the Test Notification in Firebase Console

This screenshot shows the Firebase Cloud Messaging "Compose notification" screen. The notification title was set to "greeting" and the notification text was set to "Lt Clement NZABAMWITA". The device preview on the right shows how the notification will appear on an Android device. The "Send test message" button was used to deliver the notification directly to the emulator.

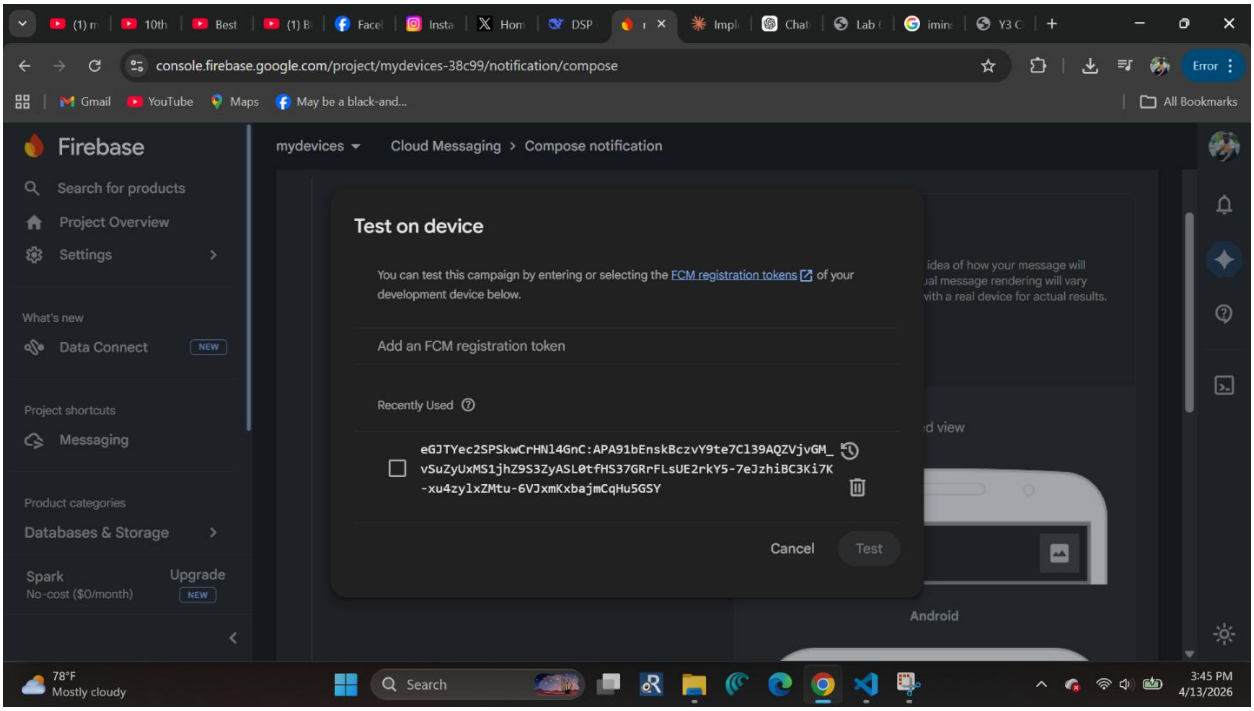

Fig. 5 — Firebase Console Compose Notification screen with message details

Screenshot 6 — Cloud Messaging Campaign History

This screenshot shows the Firebase Cloud Messaging interface with the notification campaign that was created. It confirms that the message was successfully sent from the Firebase Console. The Cloud Messaging section is accessible under DevOps & Engagement in the Firebase project sidebar.

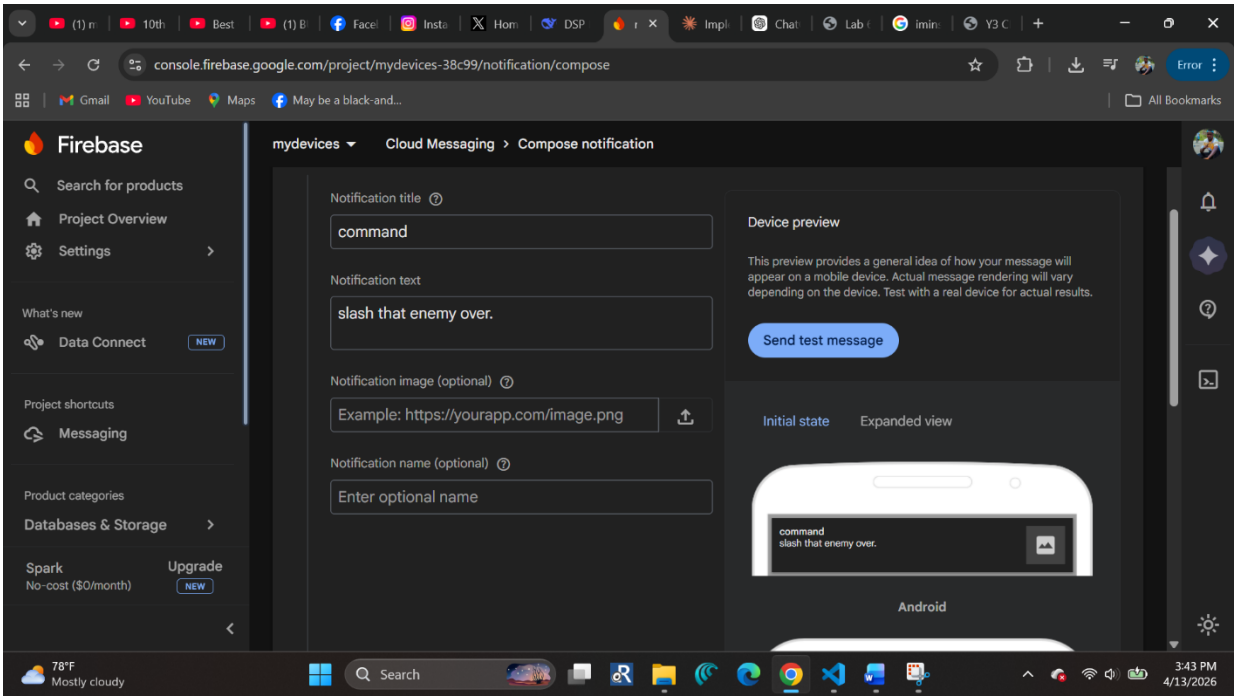

Fig. 6 — Firebase Console Cloud Messaging section

Screenshot 7 — Notification Received on Device (Foreground)

This screenshot shows the notification being received while the app is open (foreground state). Two things happened simultaneously: (1) a system notification appeared in the notification bar at the top showing "greeting __hello chalie", and (2) a popup dialog appeared inside the app with the full message content and a timestamp showing when it was received. The message history at the bottom also shows the notification was logged.

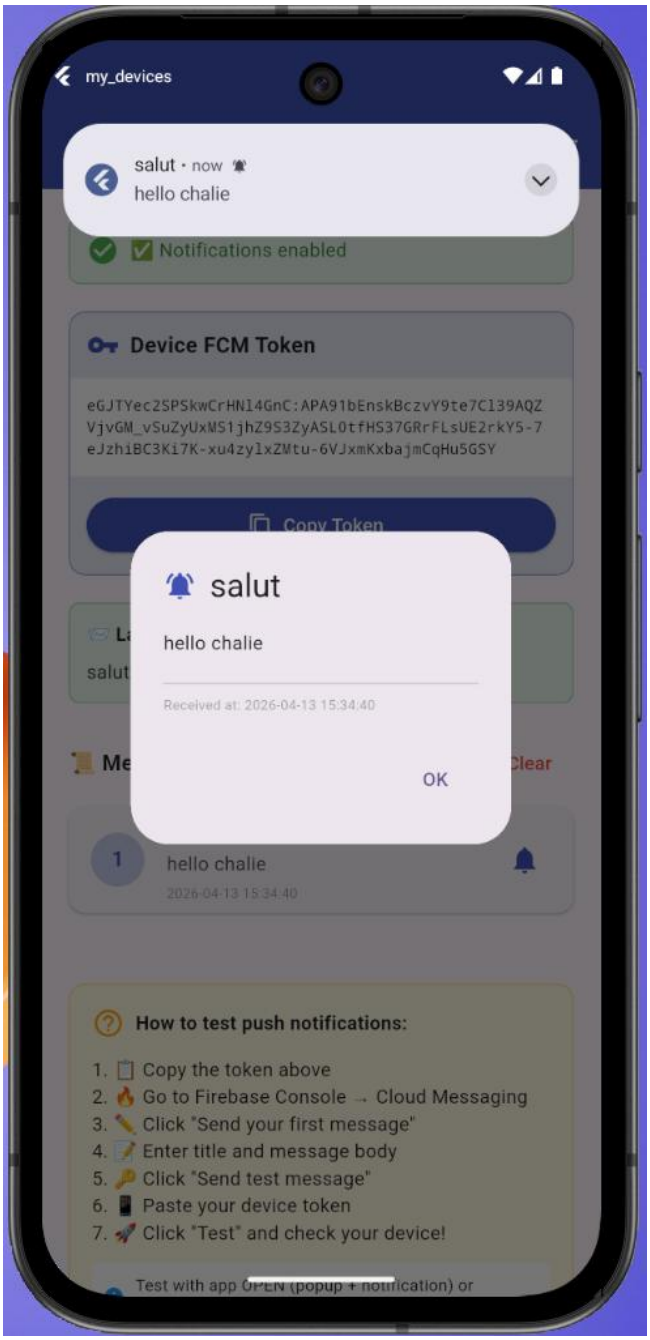

Fig. 7 — Popup dialog and system notification received while app is open

Screenshot 8 — Message History and Last Received Message

This screenshot shows the Push Notifications screen after receiving multiple notifications. The "Last Received Message" section at the top displays the most recent notification content. The "Message History" section below shows a chronological list of all received notifications with their titles, body text, and timestamps. Two notifications were received, both labeled "greeting" with the message "slash that enemy over", received at 3:41:02 and 3:41:05.

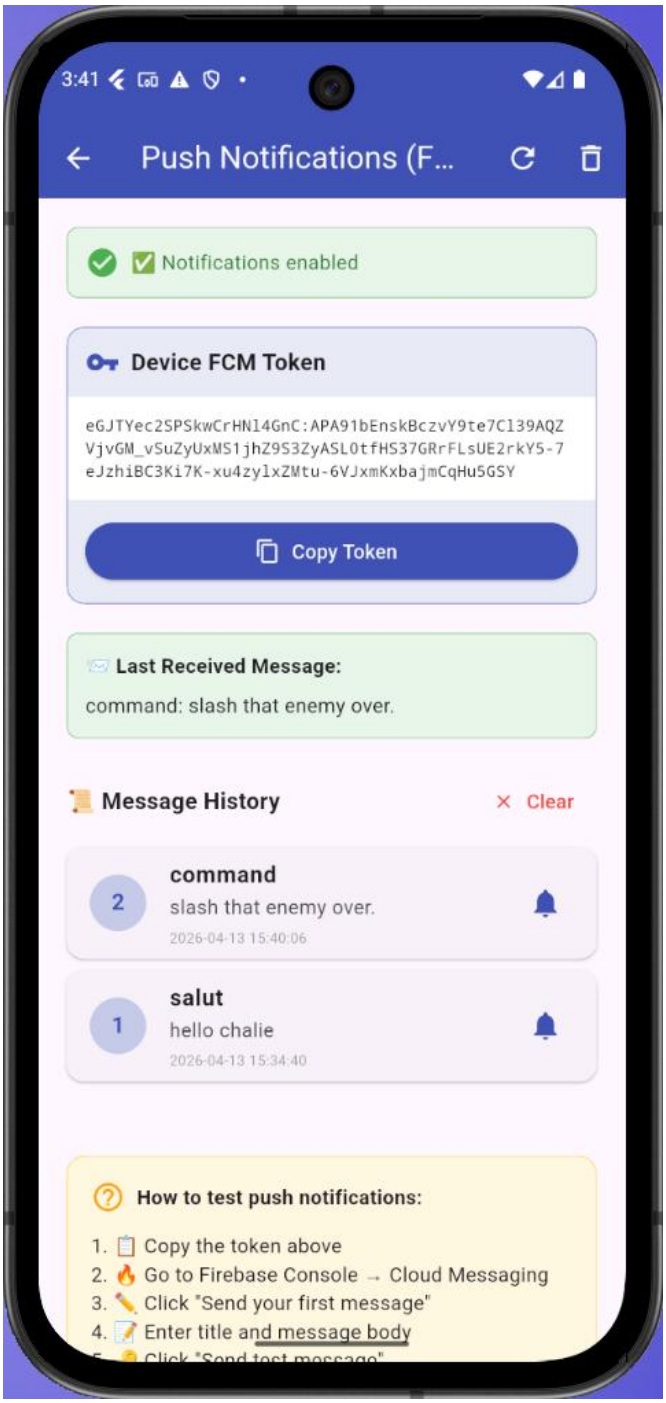

Fig. 8 — Message history showing received notifications with timestamps

3. Observations

1. Firebase was successfully integrated into the Flutter project using the google-services.json configuration file and Gradle plugin.
2. The FCM token was uniquely generated for the emulator device and remained consistent across app restarts.
3. Notifications received while the app was open (foreground) triggered both a popup dialog inside the app and a system notification in the notification bar.
4. The app correctly stored received notifications in a message history list with timestamps.
5. The emulator supports FCM push notifications, making it possible to complete the full lab without a physical device.
6. The core library desugaring fix was required in android/app/build.gradle to resolve a compatibility issue with flutter_local_notifications on Android.
7. The FirebaseMessaging.onBackgroundMessage() handler was registered in main.dart to handle messages received when the app is closed or in the background.
8. "Unable to log event: analytics library is missing" warnings appeared in the log but did not affect functionality — this is expected when firebase_analytics is not included.

3.1 HANDWRITTEN

OBSERVATIONS

1. Firebase setup complexity
- Had to download and place google-services.json in the correct folder (android/app/).
- Missing this file caused token retrieval errors.
- Required proper package name matching (com.example.my_devices).

2. Dependency version compatibility.
- multiple package had version conflicts.
- Needed to ensure firebase_core, firebase_messaging, and flutter_local_notification worked together.
- flutter clean and flutter pub get were essential after adding dependencies

3. Android manifest permisions.
- Had to add <uses-permission android:name="android.permission.POST_NOTIFICATIONS"> for android 13+

4. Syntax error with resolvedplatformspecificImplementation
- Error: used dot (.) instead of angle brackets (<>)
- fix: changed to resolvedplatformspecificImplementation<AndroidFlutterLocalNotificationPlugin>()

5. missing Clipboard import
- Error: clipboard and clipboardData not defined.
- Fix: wrapped column in SingleChildScrollView and removed spacer()
- Fix: Added import 'package:flutter/services.dart'

6. Notification behavior observations.

⇔ App Open: popup dialog appears immediately + notification + message saved to history

⇔ App Background (Home button): only notification appears, message saved when opened.

⇔ App Terminated: notification bar appears, message loads when app restarts.

7. Token generation:

- Each device installation gets a unique FCM token
- Token is required in Firebase console to send test messages.
- Token can be copied to clipboard for easy testing

8. Message Delivery Timing

*) Notifications arrived with 1-2 seconds from firebase console.

*) Some delays due to emulator network (not app issue)

*) Message acknowledgement timeout warning appeared but didn't affect delivery.

9. Firebase console Testing observations.

⇒ Testing message process

- must copy exact token from app to firebase console
- Token is 150+ characters long - copy paste is essential

⇒ Popup Dialog Implementation

- Shows notification title and body
- Includes timestamp of when message was received
- Requires ShowDialog() with proper context (had to check mounted).

⇒ Emulator vs Real device

⚠ push notification works on both but some features (like certain Firebase services) better on Real devices.

⚠ Emulator showed "skipped frames" warning but notifications still worked.

KEY STEPS FOLLOWED

1. Added Firebase dependencies to pubspec.yaml
   - firebase_core: ^3.15.2
   - firebase_messaging: ^15.2.10
   - flutter_local_notifications: ^18.0.1
2. Added google_services.json to android/app/
   - Download from firebase console after registering the app
3. Updated Android build files.
4. Adding permission to AndroidManifest.txt.
5. Created main.dart with.
   - Firebase initialization
   - Background message handle with @pragma('vm:entry-point')
   - Notification permission request.
6. Creating pushnotif screen with FCM integration.
7. Adding flutter local notifications for foreground notifications.
   - Created notification channel
   - Show local notification when app is open.
8. Fixed syntax error
   changed: AndroidFlutterLocalNotificationPlugin() to <AndroidFlutterLocalNotificationPlugin>()
9. Fixing missing import
10. Fixed layout overflow
11. Tested with Firebase Console using device token

4. Summary

The lab was completed successfully. The my_devices Flutter application was configured to receive push notifications from Firebase Cloud Messaging. The device FCM token was retrieved and displayed on screen, allowing targeted notifications to be sent from the Firebase Console directly to the emulator. The app handled foreground notifications with a popup dialog and background notifications through the system notification bar. A message history feature was implemented to display all received notifications with timestamps inside the app UI.

END